

Multidomain phenotyping of painful temporomandibular disorders in young adults from the Danish National Birth Cohort



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1 BACKGROUND & AIM

Painful TMD may extend beyond localized jaw pain. Detailed multidomain characterization combining clinical examination and questionnaire data within population-based samples remains limited^{1,2}. This study characterizes this broader profile among young adults from the Danish National Birth Cohort (DNBC).

AIM: To describe and compare clinical pain, psychosocial, headache, somatosensory, pain-modulation, and selected lifestyle characteristics between young adults with and without clinically diagnosed p-TMD, drawn from the DNBC.

2 METHODS

Young adults from DNBC^{3,4}: N=70 (p-TMD n=33 & no p-TMD n=37). Groups were comparable in age, sex, and BMI (all p>0.2)

Clinical

Diagnostic Criteria for TMD examination (DC/TMD)⁵

Questionnaires

GCPS (Graded Chronic Pain Scale)⁷; Patient Health Questionnaires; sleep; headache; diet

Somatosensory

Quantitative sensory testing (QST)⁶: pressure, thermal, pinprick, wind-up

Pain modulation

Conditioned pain modulation (CPM)⁸ with cold-water conditioning

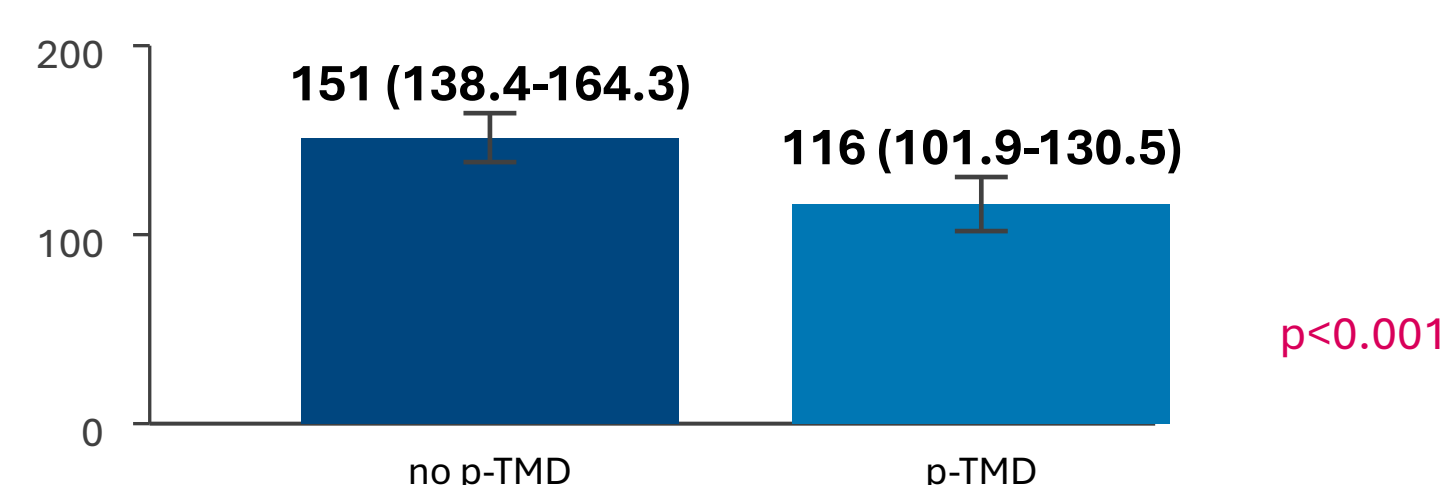
3 RESULTS

SOMATOSENSORY CHARACTERISTICS

Pressure Pain Threshold (PPT): lower threshold = greater sensitivity

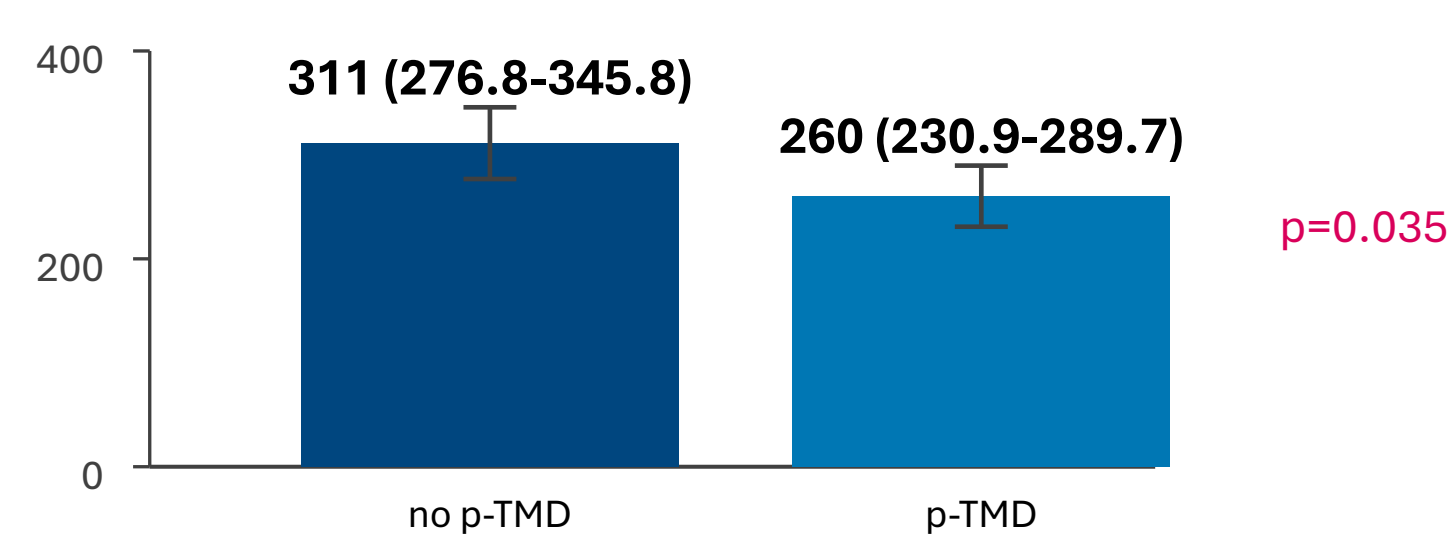
Pressure Pain Threshold (Face)

Mean (95% CI) - kPa



Pressure Pain Threshold (Hand)

Mean (95% CI) - kPa

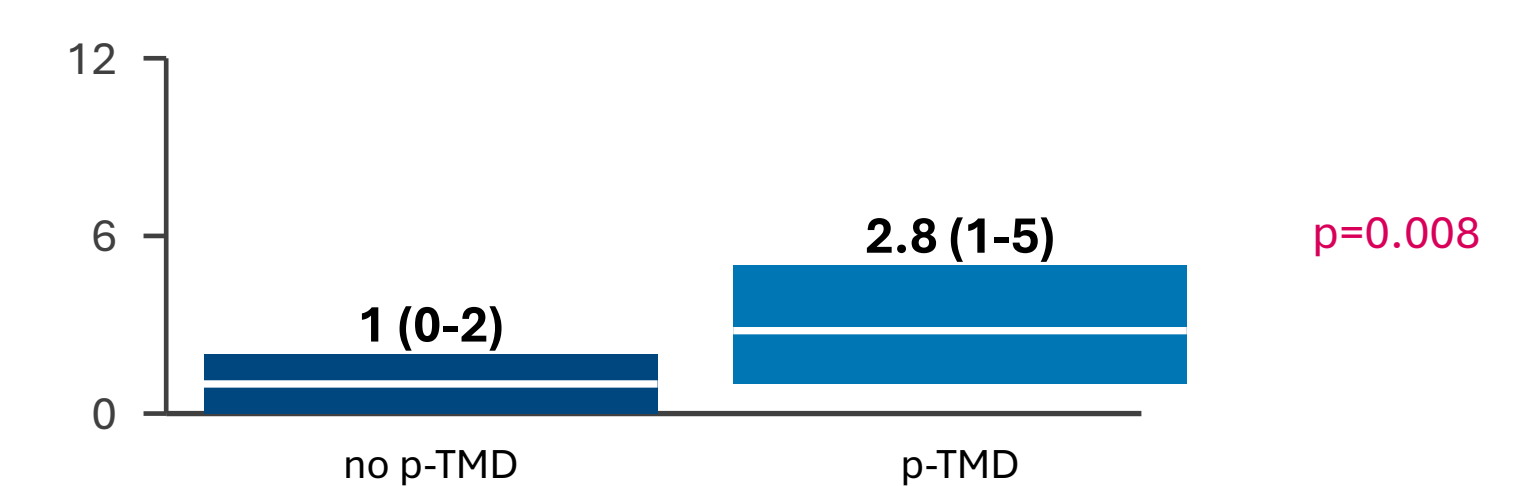


- Face wind-up ratio (WUR) and relative cold CPM also differed statistically
- No statistically significant between-group differences were observed for thermal/mechanical thresholds, hand WUR, and absolute CPM

PSYCHOSOCIAL CHARACTERISTICS

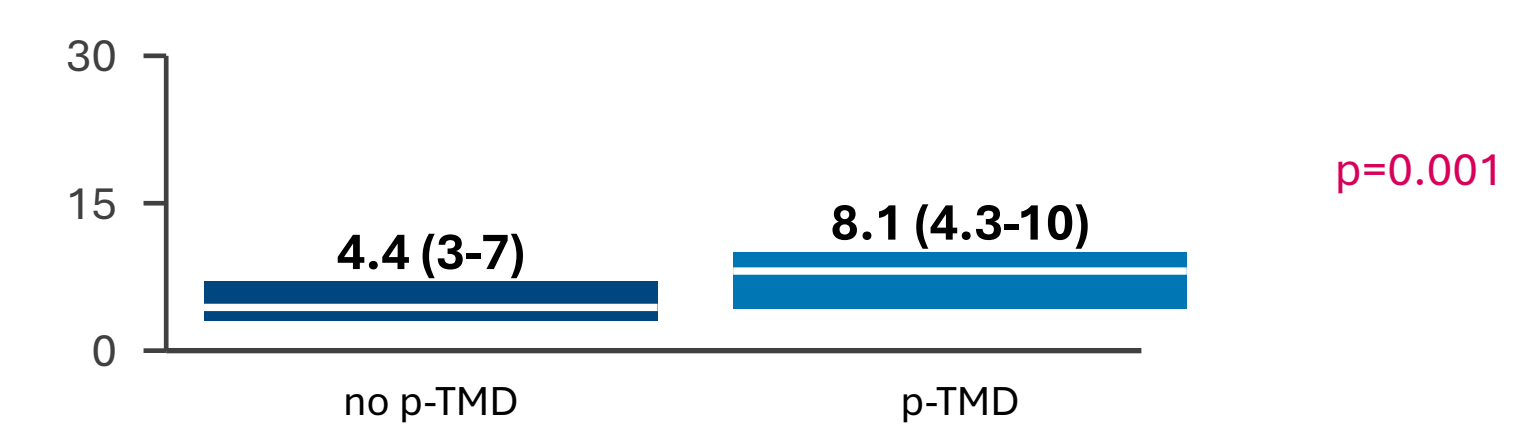
Patient Health Questionnaire (PHQ)-4⁹ (Psychological distress)

Median (IQR) - Score 0-12



PHQ-15¹⁰ (somatic burden)

Median (IQR) - Score 0-30

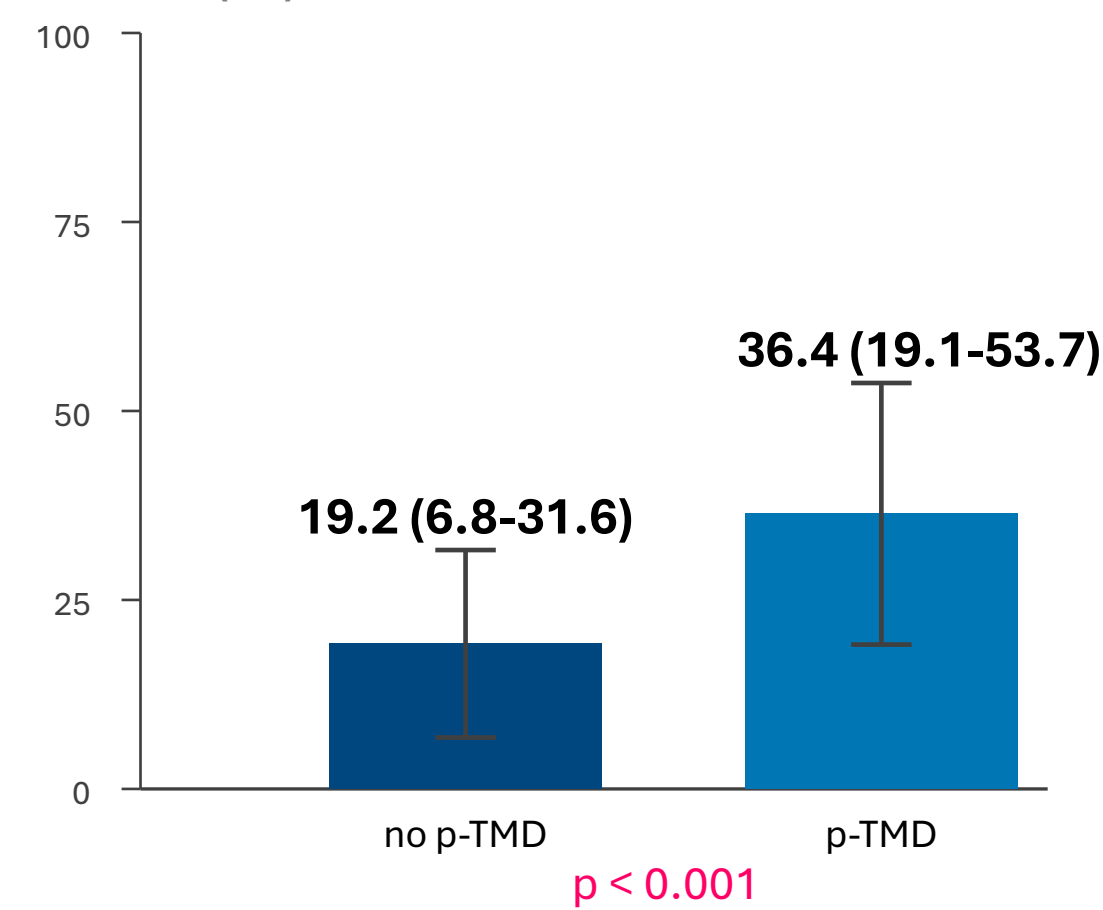


Sleep quality (Pittsburgh Sleep Quality Index)¹¹: 5.4 vs 4.4 - no significant difference (p=0.202)

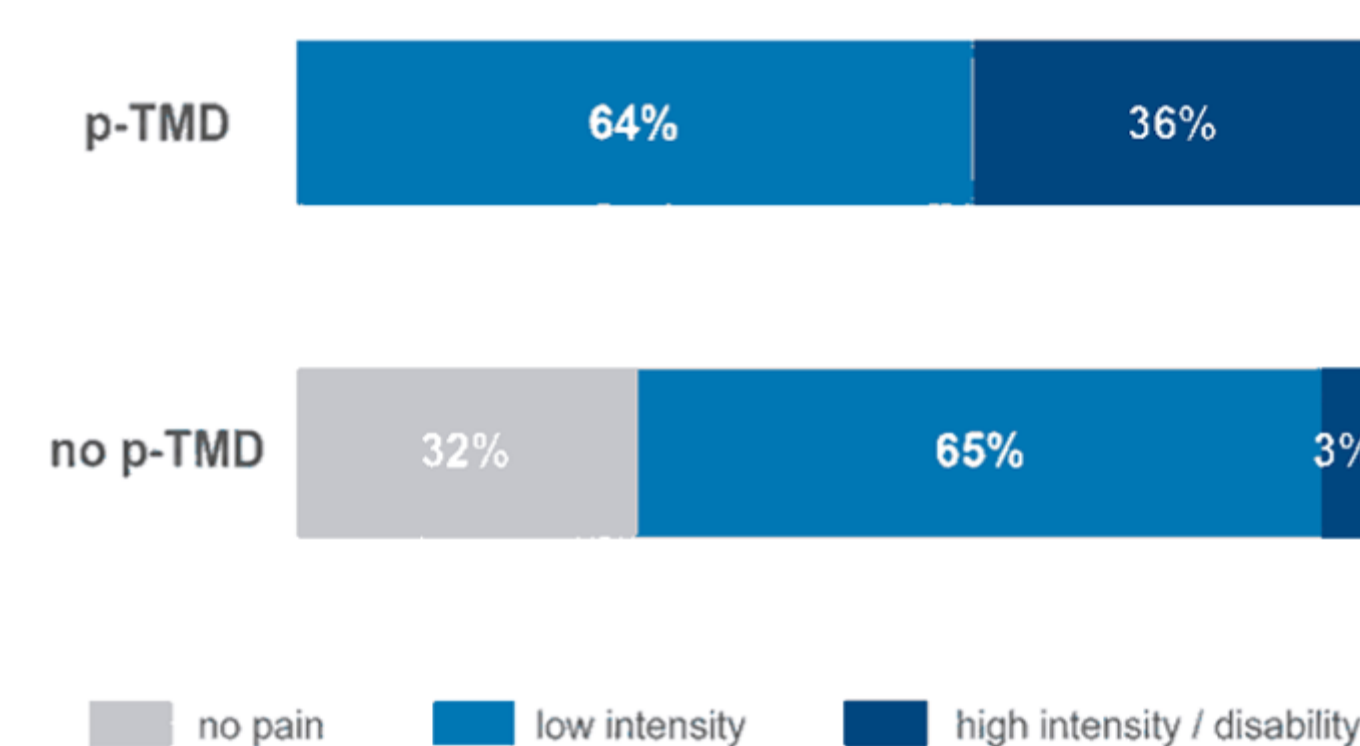
CLINICAL PAIN CHARACTERISTICS

Characteristic pain intensity

Mean (SD) - Score 0-100



Graded Chronic Pain Scale



Headache frequency: 1-14 days/month: p-TMD 26/33 (78.8%)

no p-TMD 23/37 (62.2%)

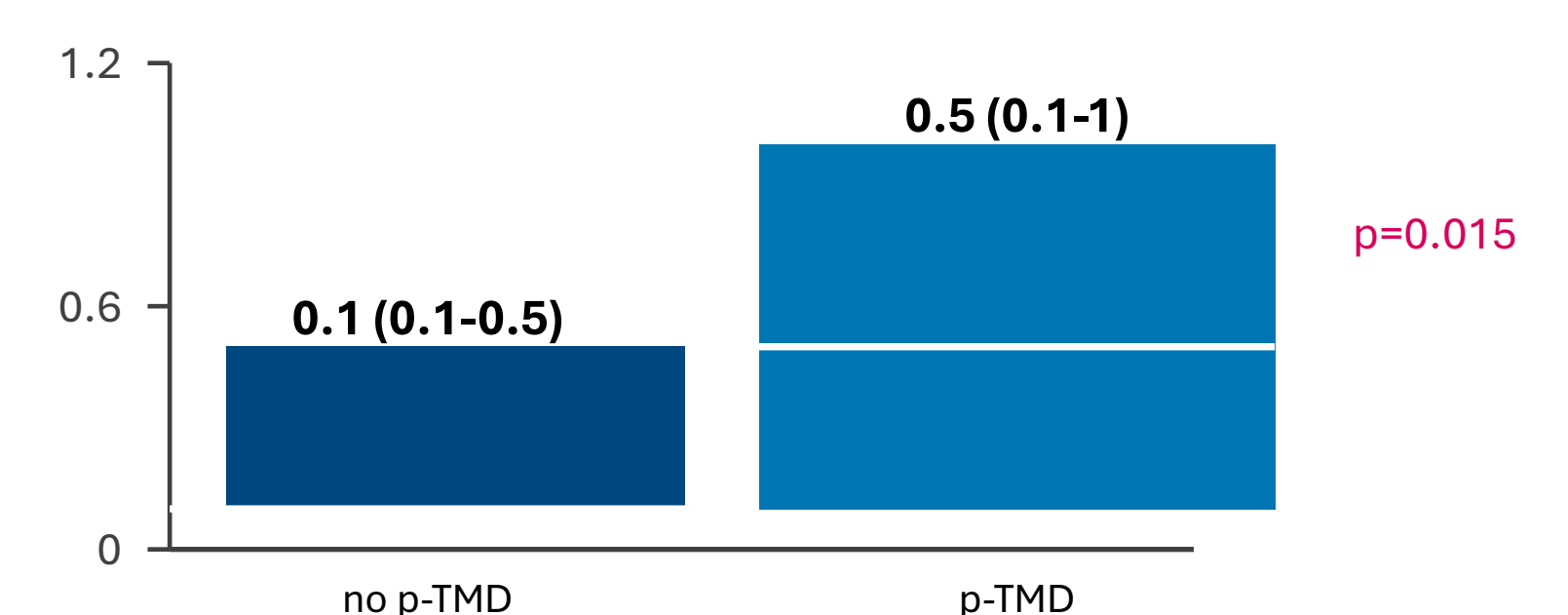
p=0.039 (overall)

Overlap: 29/33 (87.8%) in p-TMD group had ≥2 pain diagnoses

LIFESTYLE CHARACTERISTICS

Regular soda

Median (IQR) - Times/week



Food allergy/intolerance:

p-TMD 11/33 (33.3%)
no p-TMD <5/37 (<13.5%)

p=0.004

No statistically significant between-group differences were observed for:

- Smoking
- Physical activity
- Most other dietary indicators (milk, coffee, oatmeal, fruit/vegetable intake, breakfast frequency, fast food, dietary supplements)

Screener validity (TMD Pain Screener¹² vs. DC/TMD)

• Agreement (84.3%) • Sensitivity 81.8% • Specificity 86.5%

5 STRENGTHS

- Young-adult sample within a nationwide population-based cohort framework
- Standardized clinical DC/TMD classification
- Multidomain assessment extending earlier questionnaire-based DNBC research

6 LIMITATIONS

- Cross-sectional design
- No adjustment for multiple comparisons

4 CONCLUSION

Young adults with clinically classified p-TMD showed greater pain burden, pressure pain sensitivity, psychological and somatic burden, and headache frequency than those without p-TMD. The sensory pattern was selective rather than generalized. Exploratory differences in sugar-sweetened beverage intake support including lifestyle characteristics in p-TMD phenotyping, while requiring cautious interpretation

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